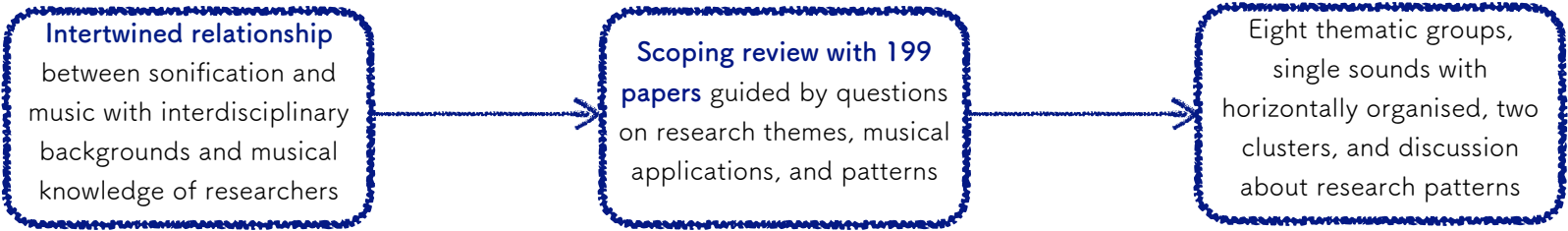


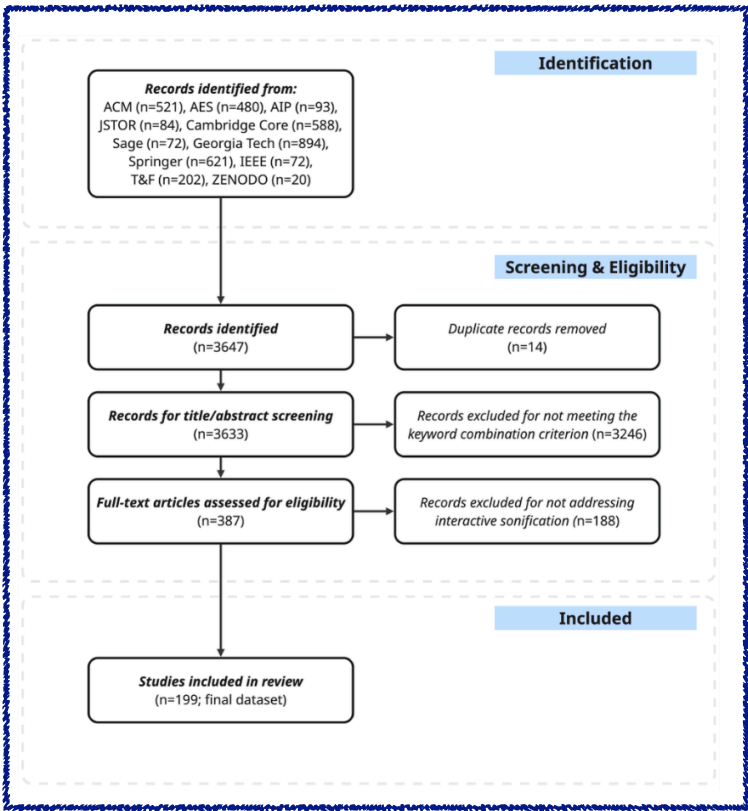
Musical Applications in Interactive Sonification: A Scoping Review

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Overview : A Structured Review of Musical Applications in Interactive Sonification for Future Research Guidance



Procedure : PRISMA-ScR



Keywords and Venues

Keywords	Search: (“Sonification” OR “Auditory Interface” OR “Auditory Display”) AND “Music” Screening: (“Sonification” OR “Auditory Interface” OR “Auditory Display” OR “Auditory Signal” OR “Auditory Representation” OR “Auditory Feedback” OR “Auditory Icon” OR “Auditory Cue/Cueing” OR “Auditory Alarm” OR “Earcon” OR “Lyricon” OR “Spearcon”) AND (“Music” OR “Rhythm” OR “Melody” OR “Harmony”)
Databases	Zenodo, ACM, Springer, Sage, IEEE Xplore, Taylor & Francis, AIP, JSTOR, Cambridge Core, Georgia Tech Library, AES E-Library
Venues	JAES, NIME, ICAD, HFES, Human Factors, Ergonomics in Design, JASA, Organised Sound, Computer Music Journal, Leonardo, LMJ

Analysis Categories

Research Theme	Art, Design, Emotion, Medicine, Movement, Nature, Space, Vision
Musical Application	Theoretical: Zero, One(horizontal), One(vertical), Two Experiential: from Performance, from Composition, For Performance, For Composition *supplemented with bridging and non-specified categories.

Thematic Groups

	Art	Design	Emotion	Medicine	Movement	Nature	Space	Vision
Single	6	47	7	9	13	7	5	19
Total	24	61	17	52	35	25	31	54
Review	1	11	0	3	1	1	4	0
Theoretical	1	9	1	1	1	0	3	2
Human	15	33	14	39	27	14	16	37

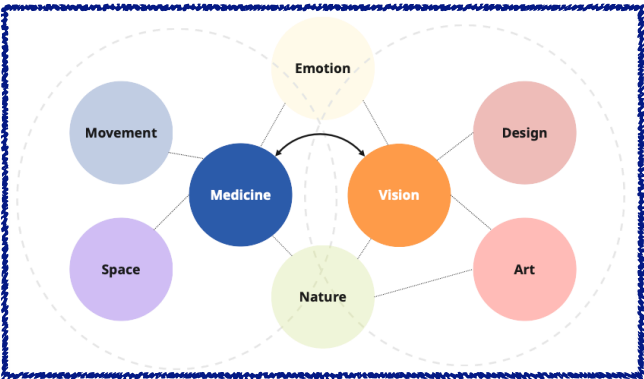
Thematic Overlaps

	Art	Design	Emotion	Medicine	Movement	Nature	Space	Vision
Art	-	0	1	3	2	7	5	7
Design	0	-	1	0	0	1	5	8
Emotion	1	1	-	4	2	2	0	3
Medicine	3	0	4	-	15	5	9	13
Movement	2	0	2	15	-	0	3	1
Nature	7	1	2	5	0	-	3	8
Space	5	5	0	9	3	3	-	6
Vision	7	8	3	13	1	8	6	-

Musical Applications

	Zero	Zero-One(hor)	Zero-One(ver)	Zero-One(both)	One(par)	One(ver)
Theoretical	17 One(both) 2	56 One(hor)-Two 5	9 One(ver)-Two 4	5 One(both)-Two 1	5 Two 16	2 Sum 122
Experiential	from Perform 1 Both Perform 4	from Comp 3 Both Comp 2	from Both 1 Sum 16	For Perform 2	For Comp 3	For Both 0
Others	T/E Both 11	Non-Specified 30	Non-T 4	Sum 45		Total 183(16)

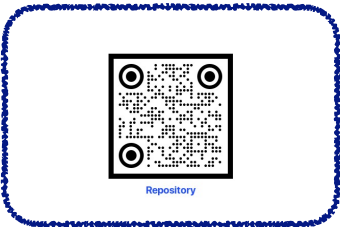
Thematic Clusters



Discussion : Research Patterns

- Thematic clusters primarily centred on **medicine and vision**.
- Music was **commonly used** to structure sonic interactions.
- Musical concepts were often **simplified from theory** or treated as **abstract references**.
- Effectiveness ultimately depended on their **appropriate application to the research context**.

Data



Paper

